

Thoughts Inspired by Bennett's "Five Breakthroughs: A First Approximation of Brain Evolution From Early Bilaterians to Humans¹."

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Humuhumunukunukuapua`a Works Special Projects ($-2 \leq \text{TRL} \leq 0.5$)

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¹Bennett, M.S., Five Breakthroughs: A First Approximation of Brain Evolution From Early Bilaterians to Humans. Frontiers in Neuroanatomy, 2021. 15.

- Definitions and basic concepts.
- Introduction.
- Goal of Bennett's model.
- Model.
 - Breakthroughs.
- Breakthrough #1
- Transition from breakthrough #1 to #2.

Definition: Phylogeny²

1. The evolutionary history of a kind of organism.
2. The evolution of a genetically related group of organisms as distinguished from the development of the individual organism.
- 3. The history or course of the development of something (such as a word or custom.)**

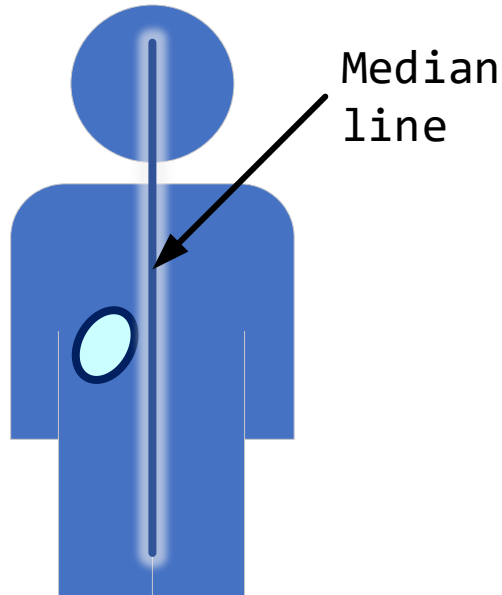
²Merriam-Webster

Definitions: Bilaterianism³

- Bilaterian: an animal having bilateral symmetry.
- Bilateral symmetry: symmetry in which similar anatomical parts are arranged on opposite sides of a median axis so that only one plane can divide the individual into essentially identical halves.

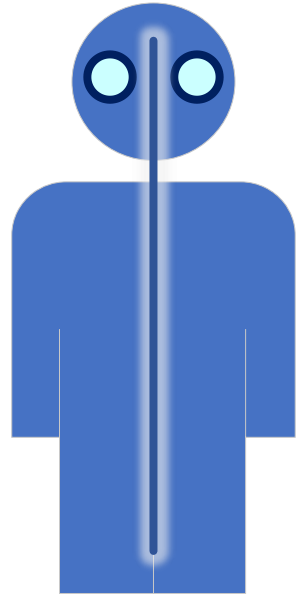
³Merriam-Webster

Basic Concept: Laterality



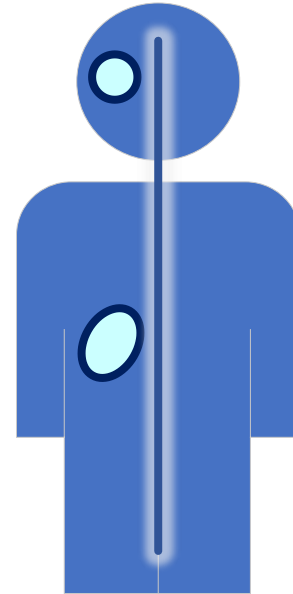
Unilateral

Present on one side only: the liver is unilateral.



Bilateral

Present on both sides: the eyes are bilateral.



Ipsilateral

Same side of body: the right eye is ipsilateral **to** the liver.

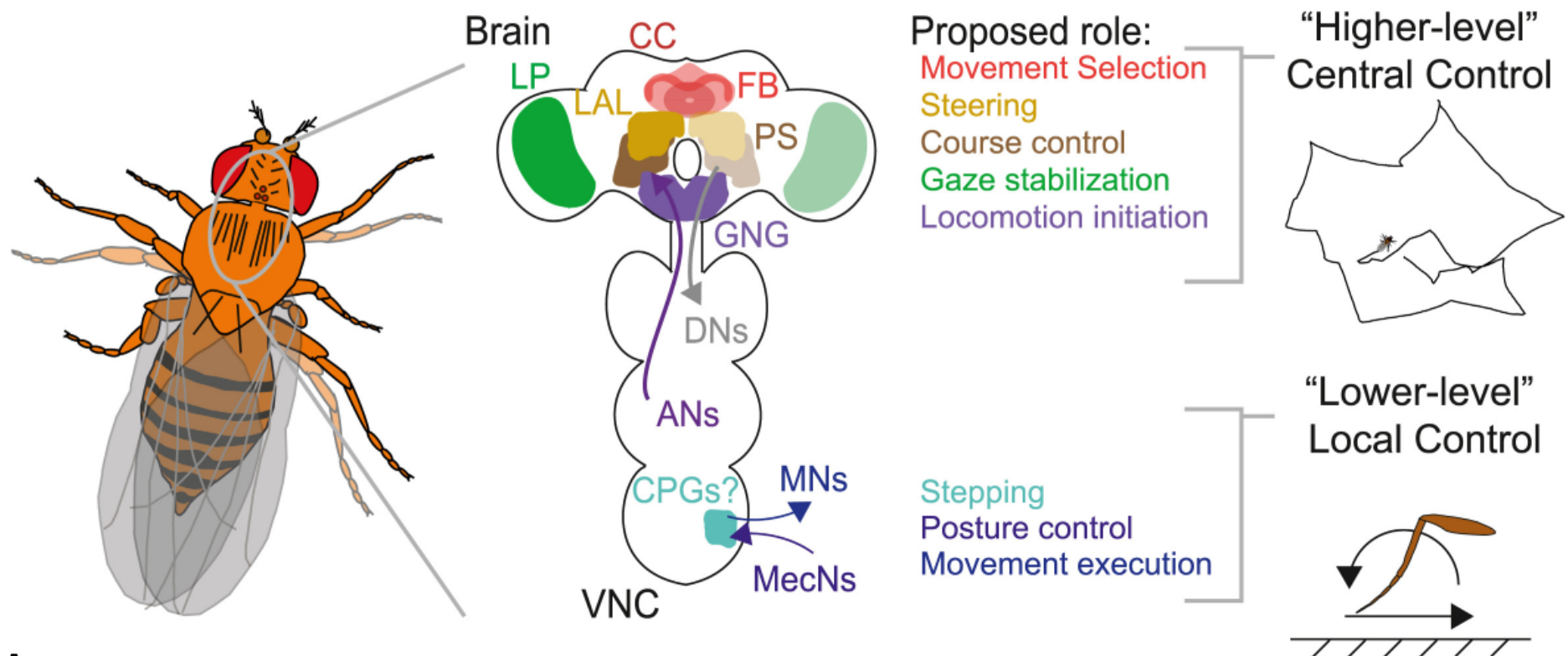


Contralateral

Opposite sides of body: the right eye is contralateral **to** the left eye.

Relative

Example Bilaterian: Fruit Fly



Acronyms

AN = Ascending Neurons

CC = Central Complex

CPG = Central Pattern Generator

FB = Fan-shaped Body

LAL = Lateral Auxiliary Lobe

LP = Lobula Plate

GNG = GNathal Ganglion

MecN = ???

MN = Motor Neuron

PS = Posterior Slope

VNC = Ventral Nerve Chord

Current Opinion in Neurobiology

Figure 1 reproduced from Cruz, T.L. and M.E. Chiappe, Multilevel visuomotor control of locomotion in *Drosophila*. Current Opinion in Neurobiology, 2023. 82.

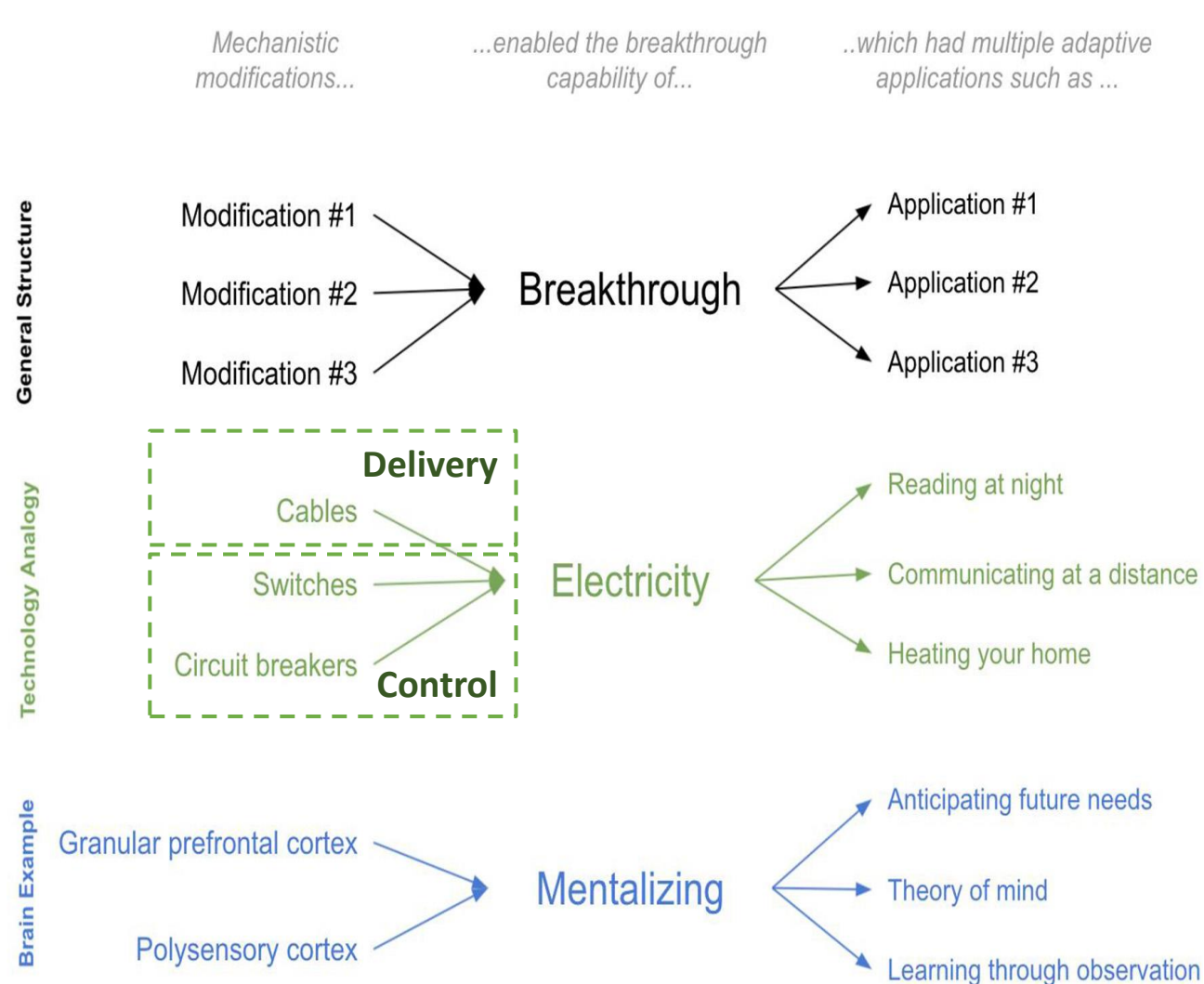
- Gain insights into the underlying mechanisms of human brain function as well as the phylogenetic origin of various features of human behavior.
- **Main thesis:**
 - *“Many neural modifications can be explained as having enabled a new single breakthrough, which was thereby applied in many adaptive ways.”*

¹Bennett, M.S., Five Breakthroughs: A First Approximation of Brain Evolution From Early Bilaterians to Humans. Frontiers in Neuroanatomy, 2021. 15.

- Employs the concept of **breakthroughs** to explain:
 - Suites of brain modifications [throughout evolution.]
 - Various adaptive behaviors that the modifications enabled.
- Notes:
 - Evolution of human brains is an example of "*progressive complexification*" that is "*elegantly articulated as phylogenetic refinement.*"
 - The value of [the] physical modifications is defined only in the context of these adaptive applications.

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Breakthrough: Exemplifications

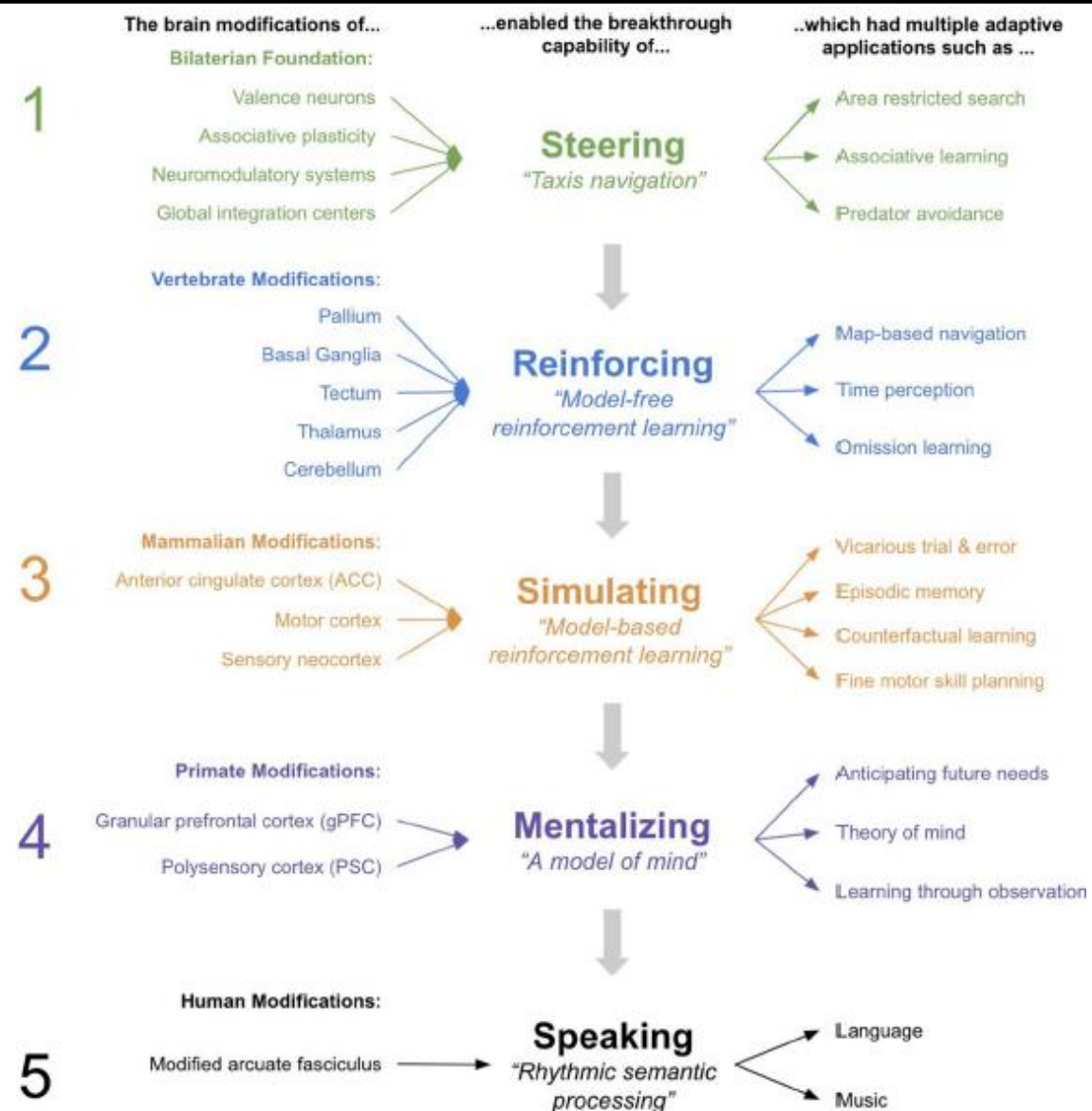


Benefits of the breakthrough-based model:

1. It enables a useful first approximation of both physical modifications and the adaptive value they provide.
2. It provides a simple interpretation of [time] ordered constraints; prior breakthroughs imposed important constraints on future breakthroughs.

Figure 1 reproduced from Bennett, M.S., Five Breakthroughs: A First Approximation of Brain Evolution From Early Bilaterians to Humans. *Frontiers in Neuroanatomy*, 2021. 15.

Five Breakthroughs



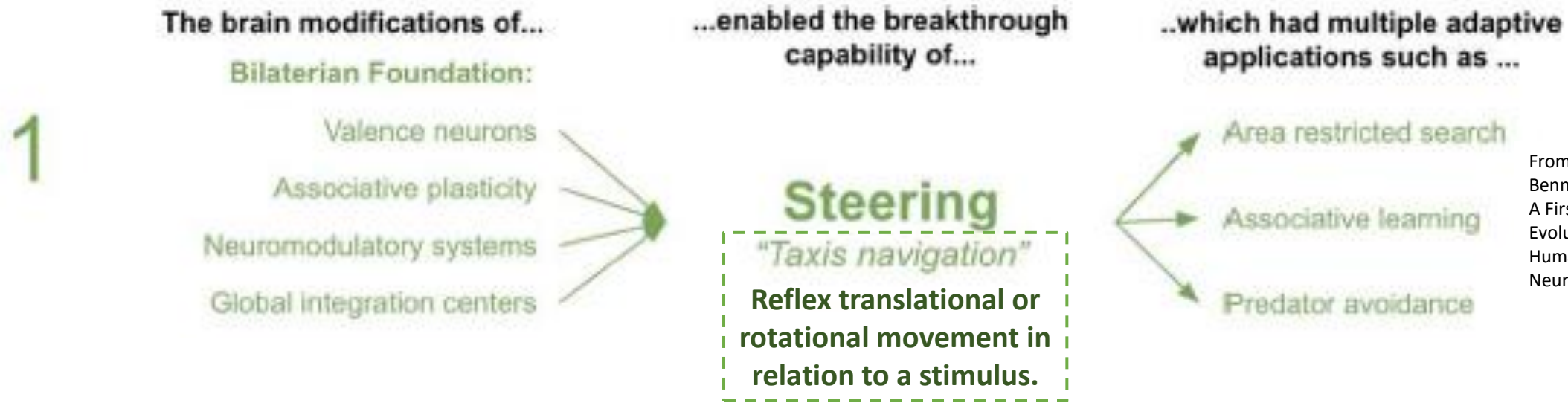
Another thesis

"Convergent evolution is not the exception but the rule."

Another definition

Convergent evolution (M-W) : the independent development of similar traits or features (as of body structure or behavior) in unrelated or distantly related species or lineages that typically occupy similar environments or ecological niches.

Breakthrough #1: Steering in Early Bilats



From Figure 7 reproduced from Bennett, M.S., Five Breakthroughs: A First Approximation of Brain Evolution From Early Bilaterians to Humans. Frontiers in Neuroanatomy, 2021. 15.

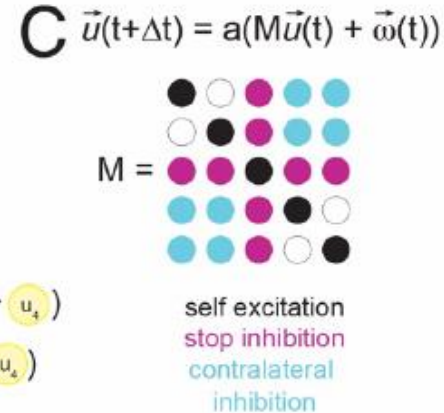
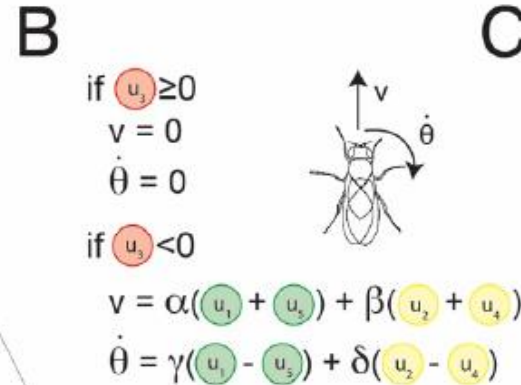
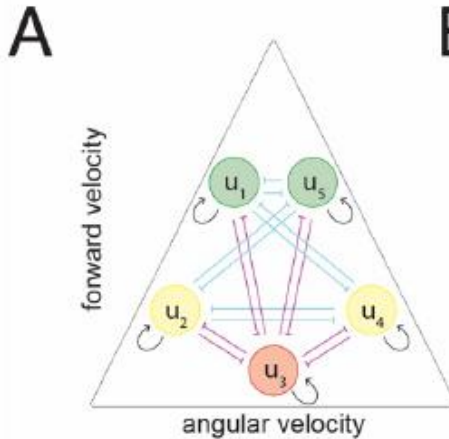
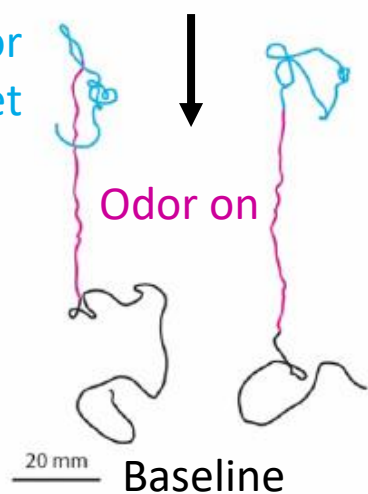
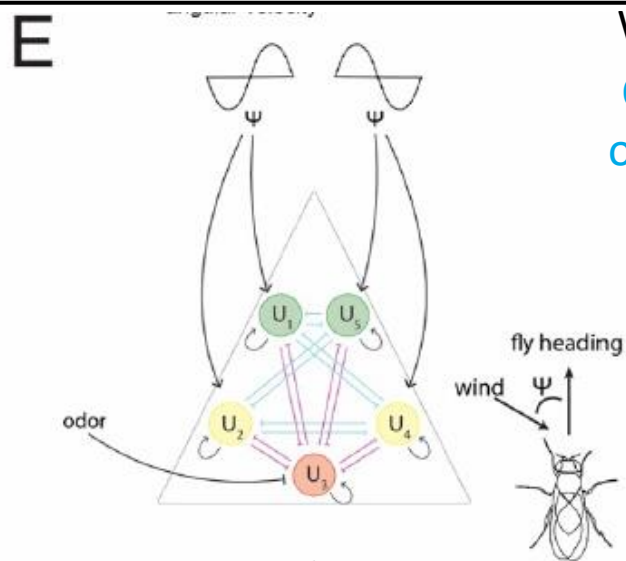
Assumes:

1. A bilateral body plan.
2. Existence of valence sensory neurons that connected to global neural integration centers (the first “brain”).
3. Neurobiological mechanisms of associative plasticity.
4. Neuromodulatory systems that generated persistent behavioral states,

Exemplification: Fruit fly (*Drosophila Melanogaster*)

Taxis model

Bogdan's model in Nengo

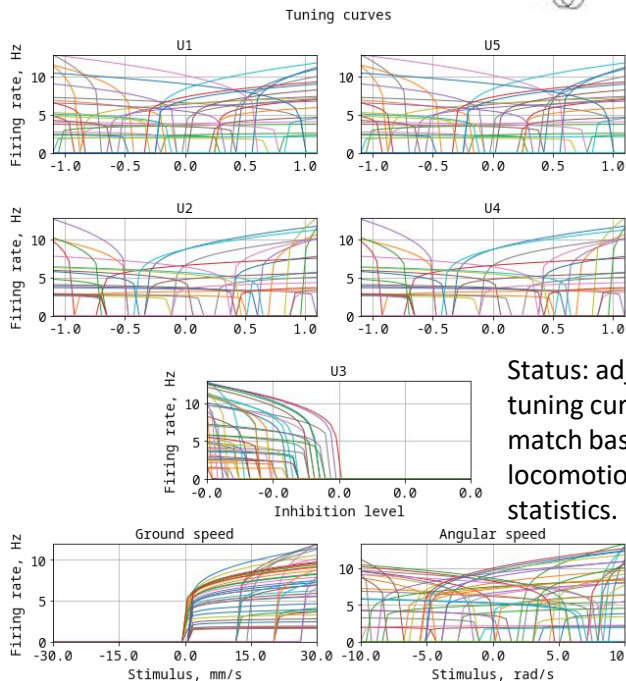
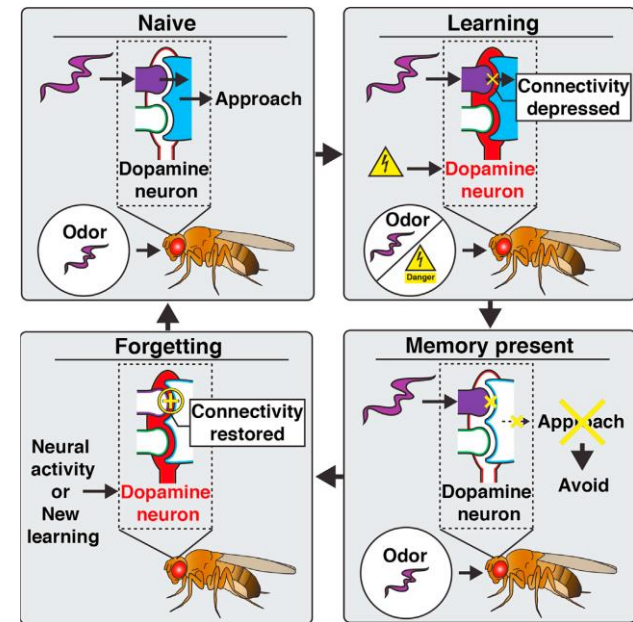


From Figures 3 and Supp 2 Gattuso, H., et al., Inhibitory control of locomotor statistics in walking *Drosophila*. 2024, bioRxiv.

Assumptions (for early bilaterians):

- ✓ A bilateral body plan.
- ✓ Existence of valence sensory neurons that connected to global neural integration centers (the first "brain").
- ✓ Neurobiological mechanisms of associative plasticity.
- ✓ Neuromodulatory systems that generated persistent behavioral states.

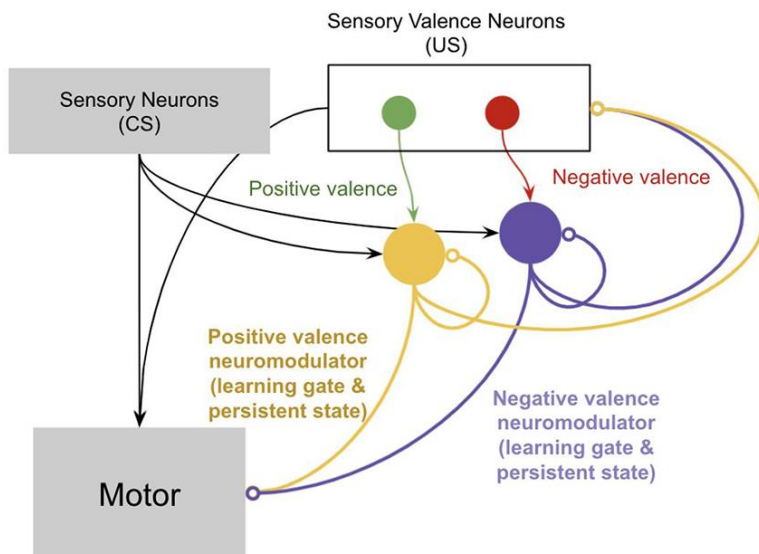
Associative plasticity?



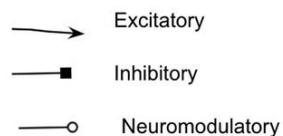
Berry, J.A., A. Phan, and R.L. Davis, Dopamine Neurons Mediate Learning and Forgetting through Bidirectional Modulation of a Memory Trace. *Cell Reports*, 2018. 25(3): p. 651-662.

From Breakthrough #1 to #2: Reinforcing

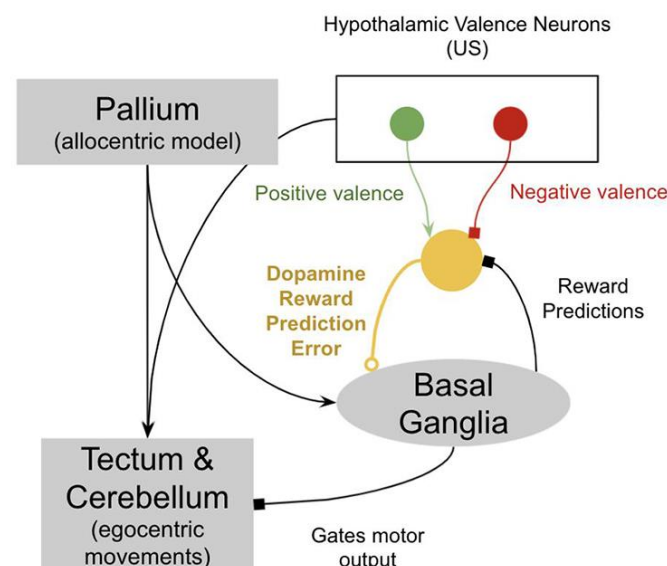
Bilaterian Associative Learning



- ✓ Classical & Instrumental Conditioning
- ✓ Blocking
- ✓ Secondary conditioning
- ✓ Overshadowing
- ✓ Latent Inhibition
- ✗ Omission Learning
- ✗ Interval Timing
- ✗ Map-Based Navigation



Vertebrate Reinforcement Learning



- ✓ Classical & Instrumental Conditioning
- ✓ Blocking
- ✓ Secondary conditioning
- ✓ Overshadowing
- ✓ Latent Inhibition
- ✓ Omission Learning
- ✓ Interval Timing
- ✓ Map-Based Navigation

Figure 3 reproduced from Bennett, M.S., Five Breakthroughs: A First Approximation of Brain Evolution From Early Bilaterians to Humans. Frontiers in Neuroanatomy, 2021. 15.

To come...